## Compiled Solutions – Intermediate Division

1. **00-01 C3 Graph Theory**

Square the adjacency matric to see that there are 13 paths of length 2. 13

$$\begin{vmatrix} 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{vmatrix}^2 = \begin{vmatrix} 1 & 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{vmatrix}$$

2. **00-01 C3 Graph Theory**

A cycle is a simple path, no vertex is repeated, except that the first and last vertices are the same. ABCA is the only path that exists. B

3. **01-02 C3 Graph Theory**

By definition a simple path is a path with no vertex repeated. Choice A is not a valid path. Choice B has vertex B repeated. Choice C is a a simple path. Choice D is not a valid path. C

4. **01-02 C3 Graph Theory**

The adjacency matrix is as follows: As shown

$$\begin{vmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{vmatrix}$$

5. **02-03 C3 Graph Theory**

A cycle is a simple path that begins and ends with the same vertex 8
The cycles are: AA, AEA, ABEA, ABCA, ADEA, ADCA, AEBCA, ADEBCA.

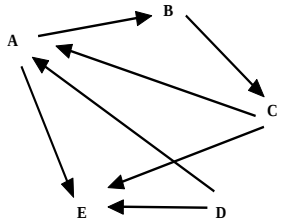
6. **02-03 C3 Graph Theory**

Square the adjacency matric to see that there are 19 paths of length 2. 19

$$\begin{vmatrix} 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{vmatrix}^2 = \begin{vmatrix} 1 & 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{vmatrix}$$

7. **03-04 C3 Graph Theory**

The directed graph is below in any form with the same edges. As shown




8. **03-04 C3 Graph Theory**

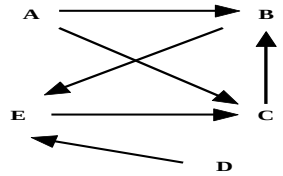
There is just one cycle in the graph: ABCA 1

9. **04-05 C3 Graph Theory**

The cycles are: ABD, ABDC, AC, ACBD, B, BDC, D 7

10. **04-05 C3 Graph Theory**

The directed graph is below in any form with the same edges. As shown




11. **05-06 C3 Graph Theory**

Square the adjacency matric to see that there are 11 paths of length 2. 11

$$\begin{vmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{vmatrix}^2 = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 1 \end{vmatrix}$$

12. **05-06 C3 Graph Theory**

The cycles are ABA, ABCA, ACBA, ACA, BCB and CC. 6

13. **06-07 C3 Graph Theory**

The cycles from vertex A are: AA, ABA, AEA, ABEA, ADEA, ADCBA, ADCBEA 7

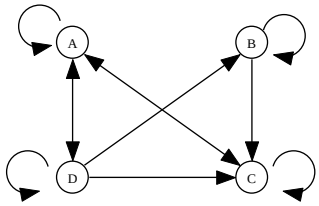
14. **06-07 C3 Graph Theory**

Square the adjacency matric to see that there are 4 paths of length 2. 4

$$\begin{vmatrix}0&0&1&1\\1&1&0&1\\1&1&0&1\\1&1&0&0\end{vmatrix}^2 = \begin{vmatrix}2&2&0&1\\2&2&1&2\\2&2&1&2\\1&1&1&2\end{vmatrix}\begin{vmatrix}0&0&1&1\\1&1&0&1\\1&1&0&1\\1&1&0&0\end{vmatrix} = \begin{vmatrix}3&3&2&4\\5&5&2&5\\5&5&2&5\\4&4&1&3\end{vmatrix}$$ 4

15. **07-08 C3 Graph Theory**

The directed graph is below in any form with the same edges. As shown




16. **07-08 C3 Graph Theory**

Square the adjacency matric to see that there are 7 paths of length 2. 7

$$\begin{vmatrix}0&1&0&1&0\\0&1&1&1&0\\0&1&0&1&0\\0&0&1&0&1\\1&0&0&0&0\end{vmatrix}^2 = \begin{vmatrix}0&1&2&1&1\\0&2&2&2&1\\0&1&2&1&1\\1&1&0&1&0\\0&1&0&1&0\end{vmatrix}$$

17. **08-09 C3 Graph Theory**

Adding the entries in the second row gives the paths from B of length 3. 8

$$\begin{vmatrix}0&1&0&1\\0&0&1&1\\1&0&1&1\\0&1&0&0\end{vmatrix}^2 = \begin{vmatrix}0&1&1&1\\1&1&1&1\\1&2&1&2\\0&0&1&1\end{vmatrix}\begin{vmatrix}0&1&0&1\\0&0&1&1\\1&0&1&1\\0&1&0&0\end{vmatrix} = \begin{vmatrix}1&1&2&2\\1&2&2&3\\1&3&3&4\\1&1&1&1\end{vmatrix}$$

18. **08-09 C3 Graph Theory**

The cycles are: AA, ABA, ACA, ACBA, BB. 5

19. **09-10 C3 Graph Theory** **6**

| | | | | |
|---|---|---|---|---|
| 0 | 1 | 1 | 1 | 1 |
| 0 | 0 | 1 | 0 | 0 |
| 0 | 1 | 0 | 0 | 1 |
| 1 | 1 | 0 | 0 | 0 |
| 0 | 0 | 0 | 1 | 0 |

3 =

| | | | | |
|---|---|---|---|---|
| 1 | 3 | 3 | 2 | 2 |
| 0 | 0 | 1 | 1 | 0 |
| 1 | 2 | 0 | 0 | 1 |
| 1 | 3 | 1 | 1 | 2 |
| 0 | 1 | 2 | 1 | 1 |

Cubing an adjacency matrix produces all the paths of length 3. 6 are 0.

20. **09-10 C3 Graph Theory**

Cycles are: CAC, CABC, CABDC, CADC, CADBC, CBC, CBDC, CDBC, CDC 9

21. **10-11 C3 Graph Theory**

The adjacency matrix is: As shown

$$\begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

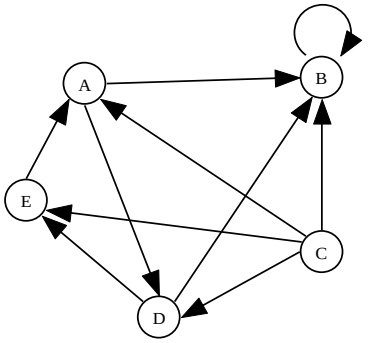
22. **10-11 C3 Graph Theory**

Adding the entries gives the paths of length 3. 32

$$\begin{vmatrix} 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{vmatrix}^2 = \begin{vmatrix} 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \end{vmatrix} \begin{vmatrix} 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{vmatrix} = \begin{vmatrix} 3 & 3 & 3 & 2 \\ 2 & 3 & 2 & 2 \\ 2 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 \end{vmatrix}$$

23. **11-12 C3 Graph Theory**

The graph is as follows: As shown



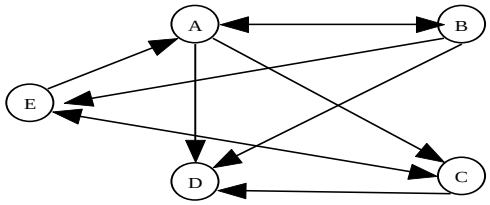

24. **11-12 C3 Graph Theory**

To find the number of paths of length 2, add the entries in the second matrix. 20

$$\begin{vmatrix}1&1&1&1\\1&0&1&0\\0&1&0&1\\0&0&1&0\end{vmatrix}^2 = \begin{vmatrix}2&2&3&2\\1&2&1&2\\1&0&2&0\\0&1&0&1\end{vmatrix}$$

25. **12-13 C3 Graph Theory**

There are no directed edged between vertices B&C and D&E. 2



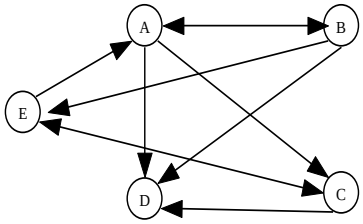

26. **12-13 C3 Graph Theory**

To find the number of paths of length 3, add the entries in the matrix cubed. 64

$$\begin{vmatrix}1&1&1&1\\1&1&1&1\\1&0&0&0\\0&0&1&0\end{vmatrix}^2 = \begin{vmatrix}3&2&3&2\\3&2&3&2\\1&1&1&1\\1&0&0&0\end{vmatrix}\begin{vmatrix}1&1&1&1\\1&1&1&1\\1&0&0&0\\0&0&1&0\end{vmatrix} = \begin{vmatrix}8&5&7&5\\8&5&7&5\\3&2&3&2\\1&1&1&1\end{vmatrix}$$

27. **13-14 C3 Graph Theory**

The graph has the following adjacency matrix: As shown




$$\begin{vmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{vmatrix}$$

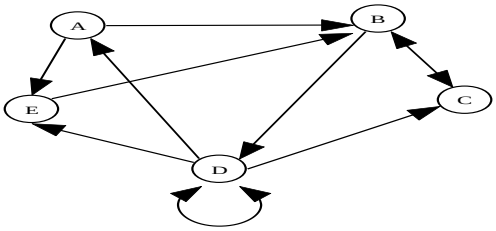
28. **13-14 C3 Graph Theory**

To find the number of paths of length 2, add the entries in the second matrix. 23

$$\begin{vmatrix} 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{vmatrix}^2 = \begin{vmatrix} 0 & 1 & 1 & 2 & 1 \\ 0 & 2 & 0 & 2 & 1 \\ 0 & 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{vmatrix}$$

29. **14-15 C3 Graph Theory**

The directed graph can be drawn in any way that has the same edges. As shown




As shown

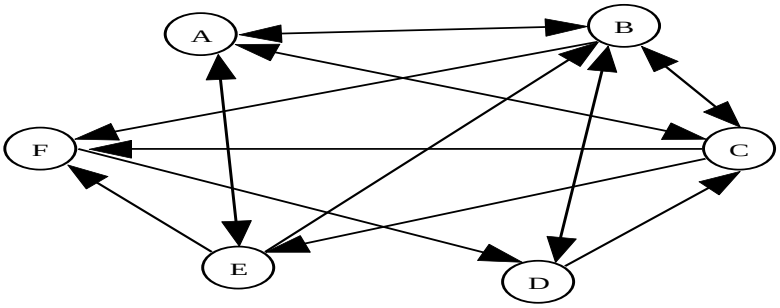
30. **14-15 C3 Graph Theory**

Vertex D has 8 paths starting there. D

$$\begin{vmatrix} 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 \end{vmatrix}^2 = \begin{vmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 3 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 0 \end{vmatrix}$$

31. **15-16 C3 Graph Theory**

The graph is shown below with edges drawn any way possible. As shown




32. **15-16 C3 Graph Theory**

There are no paths of length 3 from these vertices: AB, CB, DB AB, CB, DB

$$\begin{vmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \end{vmatrix}^2 = \begin{vmatrix} 1 & 0 & 2 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{vmatrix} \begin{vmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \end{vmatrix} = \begin{vmatrix} 2 & 0 & 3 & 2 \\ 2 & 1 & 1 & 1 \\ 2 & 0 & 2 & 1 \\ 1 & 0 & 2 & 1 \end{vmatrix}$$